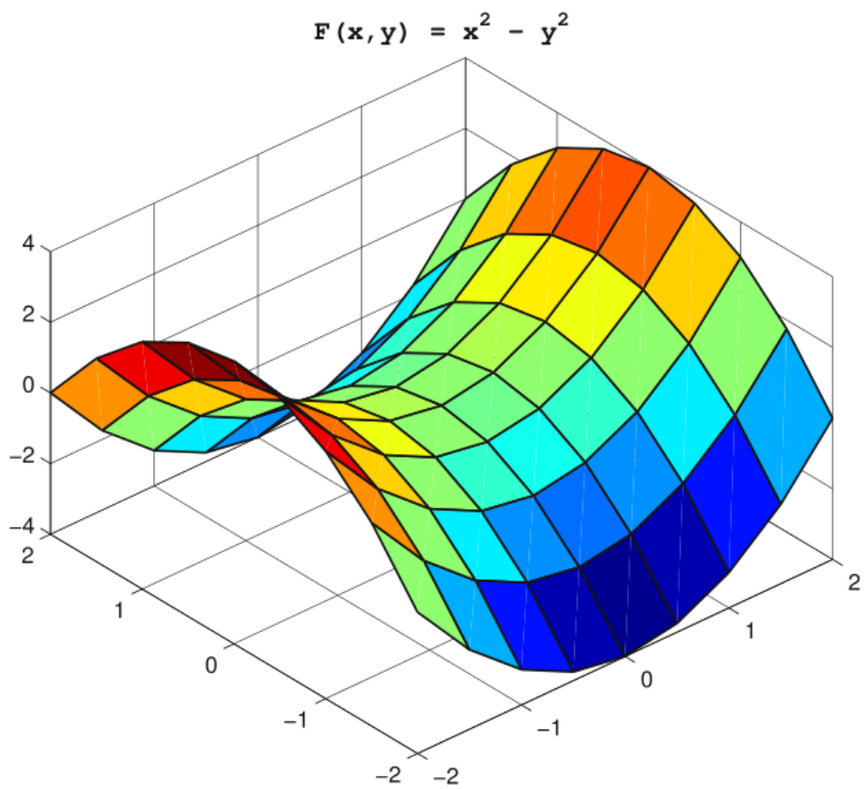
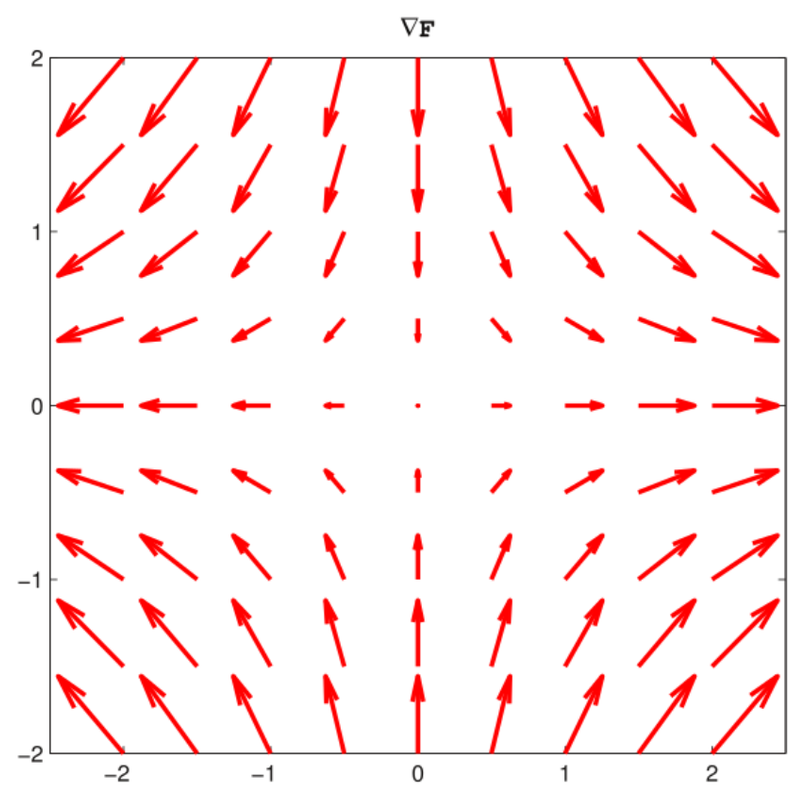
Crash Course/Refresher in Vector Calculus





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Part 1: Univariate Differentiation

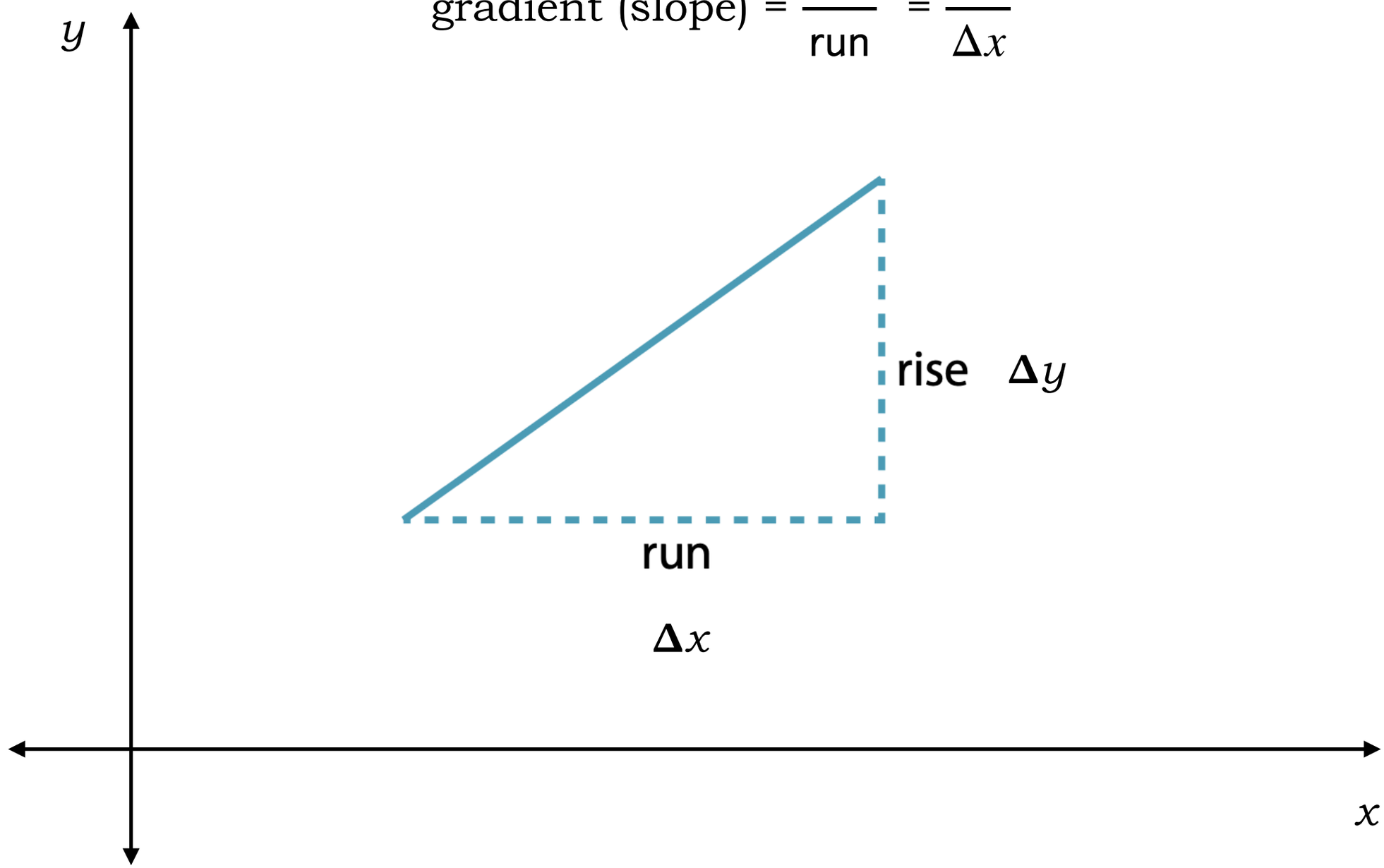
Gradients And Function Extrema

References

- *Introduction to Differential Calculus*, Australian Mathematics Sciences Institute (AMSI)
- *Thomas' Calculus in SI Units*, 13th Ed., Pearson 2016
- Many other books on Calculus

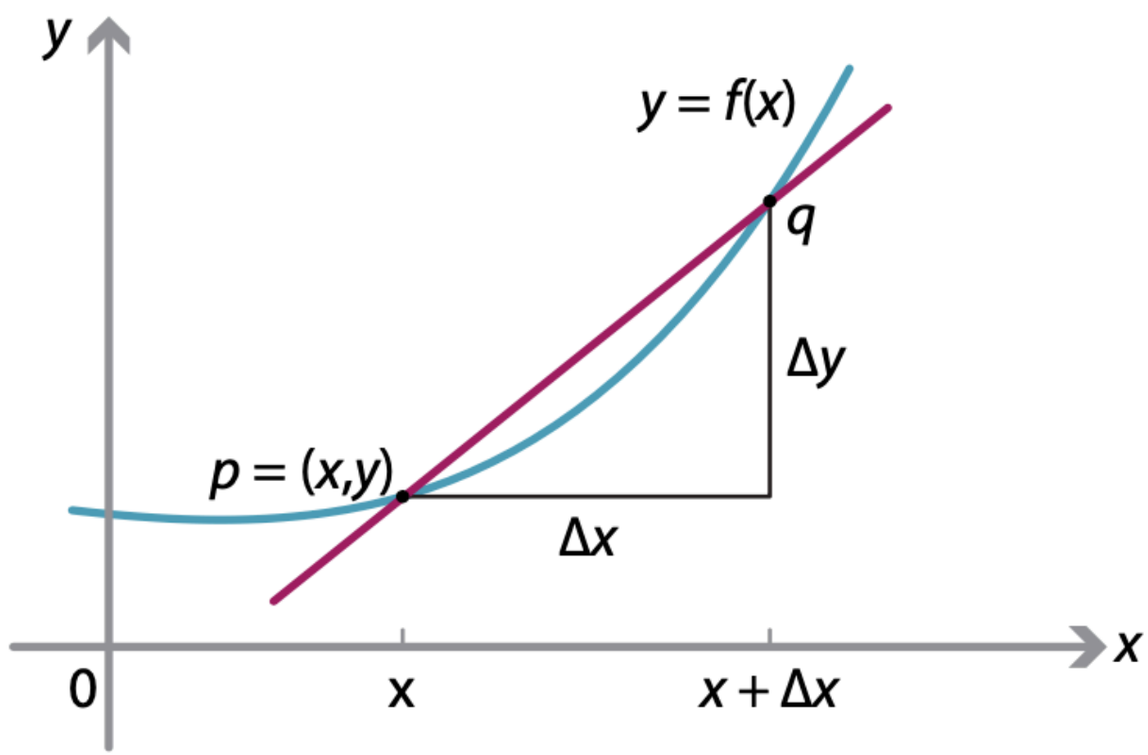
Recall Gradient of a Line

$$\text{gradient (slope)} = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x}$$



Images in this section: Introduction to Differential Calculus, Australian Mathematics Sciences Institute (AMSI)

Approximate Gradient of a Curve



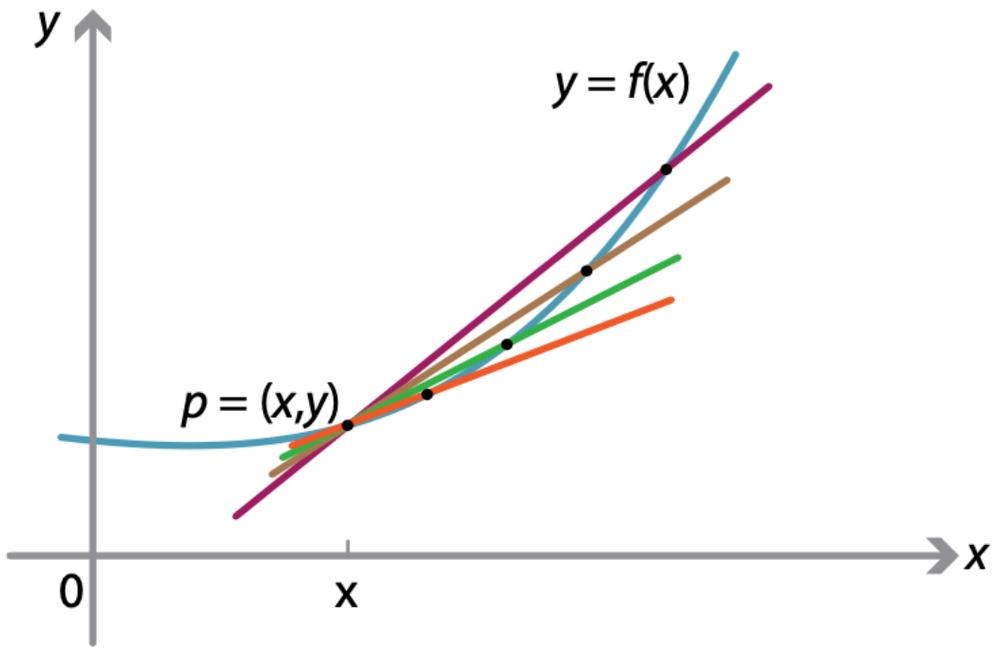
Secant connecting points on the graph $y = f(x)$ at x and $x + \Delta x$.

$$\text{slope approximated by } \frac{\Delta y}{\Delta x} = \frac{f(x+\Delta x) - f(x)}{(x+\Delta x) - x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Gradient

As

- $\Delta x \to 0$
- $q \to p$
- secant approaches tangent
- $\dfrac{\Delta y}{\Delta x} \to$ gradient (slope of tangent)



Secants on $y = f(x)$ approaching the tangent line at x.

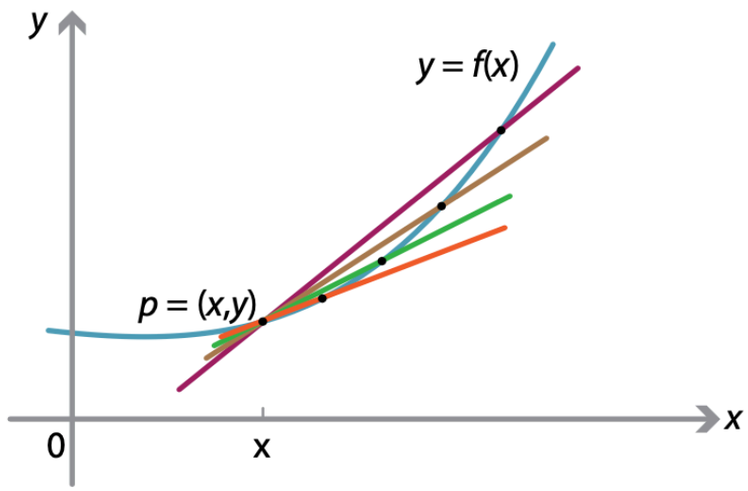
The gradient of the tangent at point p is

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

Gradient

Example

- $f(x) = x^2$



Secants on $y = f(x)$ approaching the tangent line at x.

Secants of the parabola $f(x) = x^2$

Secant between points	Δx	$\Delta y = f(x+\Delta x) - f(x)$	Gradient of secant $\frac{\Delta y}{\Delta x}$
$x = 1, \quad x = 3$	2	8	4
$x = 1, \quad x = 2$	1	3	3
$x = 1, \quad x = 1.5$	0.5	1.25	2.5
$x = 1, \quad x = 1.1$	0.1	0.21	2.1
$x = 1, \quad x = 1.001$	0.001	0.002001	2.001

$\rightarrow 2$

Derivatives

We would like a *function* that gives us the gradient for **any** x

➡ ***derivative***

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

- If $y = f(x)$ can also write as

$$f'(x) = \frac{dy}{dx}$$

- Other notations

$$\frac{d}{dx}(f(x)), \quad \frac{d}{dx}f(x), \quad \frac{d}{dx}y$$

Calculating Derivatives

Example (direct from formula)

- $f(x) = x^2$

$$\lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{(x+\Delta x)^2 - x^2}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{x^2 + 2x(\Delta x) + (\Delta x)^2 - x^2}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \left(2x + \Delta x\right) = 2x.$$

(at $x=1$, $f'(x)=2$ as before)

Derivatives of Polynomials

In fact, it is easy to show that

- **derivative of a power**

$$\frac{d}{dx}x^n = nx^{n-1}$$

It is also easy to show that:

- **derivative of a constant multiple**

$$\frac{d}{dx}cf(x) = c\frac{d}{dx}f(x)$$

- **derivative of a sum**

$$\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$$

By combining these “rules” (actually theorems) we can differentiate any polynomial…

Derivatives of Polynomials

Example

- $f(x) = x^3 + 5x^2$

$$
\begin{aligned}
f'(x) &= \frac{d}{dx}\left(x^3 + 5x^2\right) \\
&= \frac{d}{dx}\left(x^3\right) + \frac{d}{dx}\left(5x^2\right) && \text{(derivative of a sum)} \\
&= \frac{d}{dx}\left(x^3\right) + 5\,\frac{d}{dx}\left(x^2\right) && \text{(derivative of a constant multiple)} \\
&= 3x^2 + 10x && \text{(derivative of } x^n\text{).}
\end{aligned}
$$

Other functions don't fall out quite as easily as addition…

Product Rule

- **derivative of a product**

$$\frac{d}{dx}(f(x)g(x)) = f(x)\frac{d}{dx}g(x) + g(x)\frac{d}{dx}f(x)$$

Example

- $f(x) = (x^3 + 2)(x^2 + 1)$

$$f'(x) = (x^3+2)\frac{d}{dx}\left(x^2+1\right) + (x^2+1)\frac{d}{dx}\left(x^3+2\right)$$

$$= (x^3+2)\cdot 2x + (x^2+1)\cdot 3x^2$$

$$= 5x^4 + 3x^2 + 4x.$$

Exercise

- Expand f and then take the derivative to check the result is the same.

Chain Rule

This rule will be very important for us!

- **derivative of a composition of two functions (chain rule)**

$$\frac{d}{dx} f(g(x)) = \frac{d}{dg(x)} f(g(x)) \cdot \frac{d}{dx} g(x)$$

To simplify the exposition a little, define $u = g(x)$ and $y = f(g(x)) = f(u)$. Then we can write the chain rule as

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Exercise

- differentiate $f(x) = \boxed{(\underbrace{x^7 - x^2}_{u})^{42}}^{\,y}$

(Quotient Rule)

There is also a “quotient rule"

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)f'(x) - f(x)g'(x))}{(g(x))^2}$$

but I never bothered remembering this, since

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{d}{dx}\left(f(x)(g(x))^{-1}\right)$$

and we know we know from the chain rule that

$$\frac{d}{dx}\left((g(x))^{-1}\right) = -(g(x))^{-2}\frac{d}{dx}g(x)$$

Exercise

differentiate $f(x) = \dfrac{x}{\sqrt{x^2+1}}$

Exponential Functions

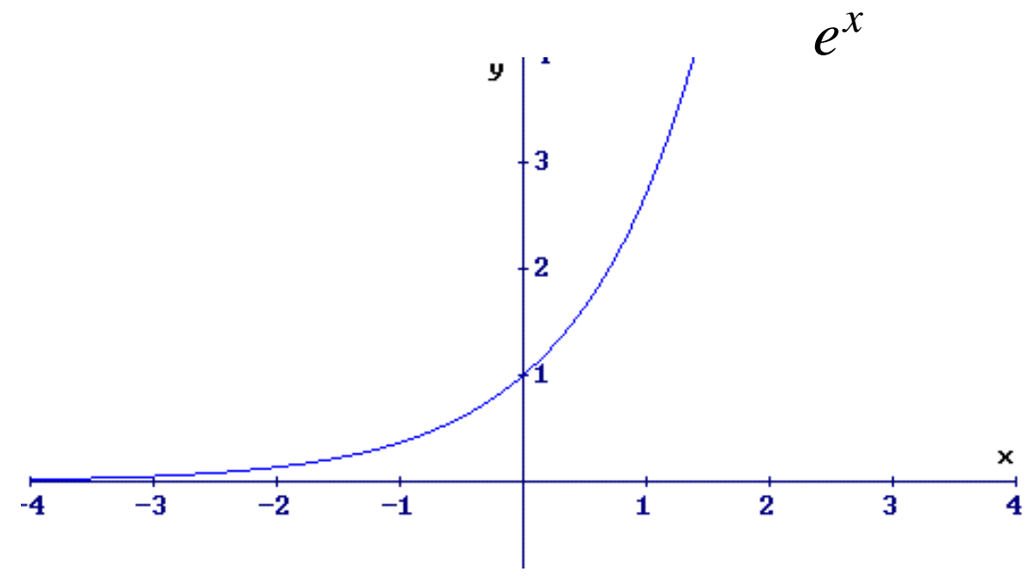
This will also be important for us!

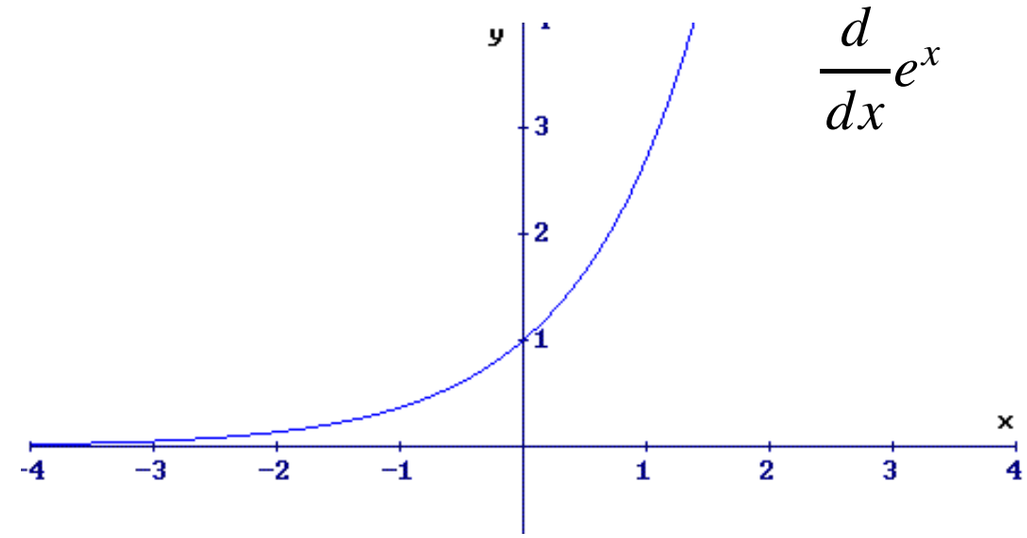
- **derivatives of exponentials**

$$\frac{d}{dx}(e^x) = e^x \qquad \text{(amazing!)}$$

Also note

$$\frac{d}{dx}(a^x) = ln(a) \cdot a^x$$





Why are Derivatives so Useful for Optimisation?

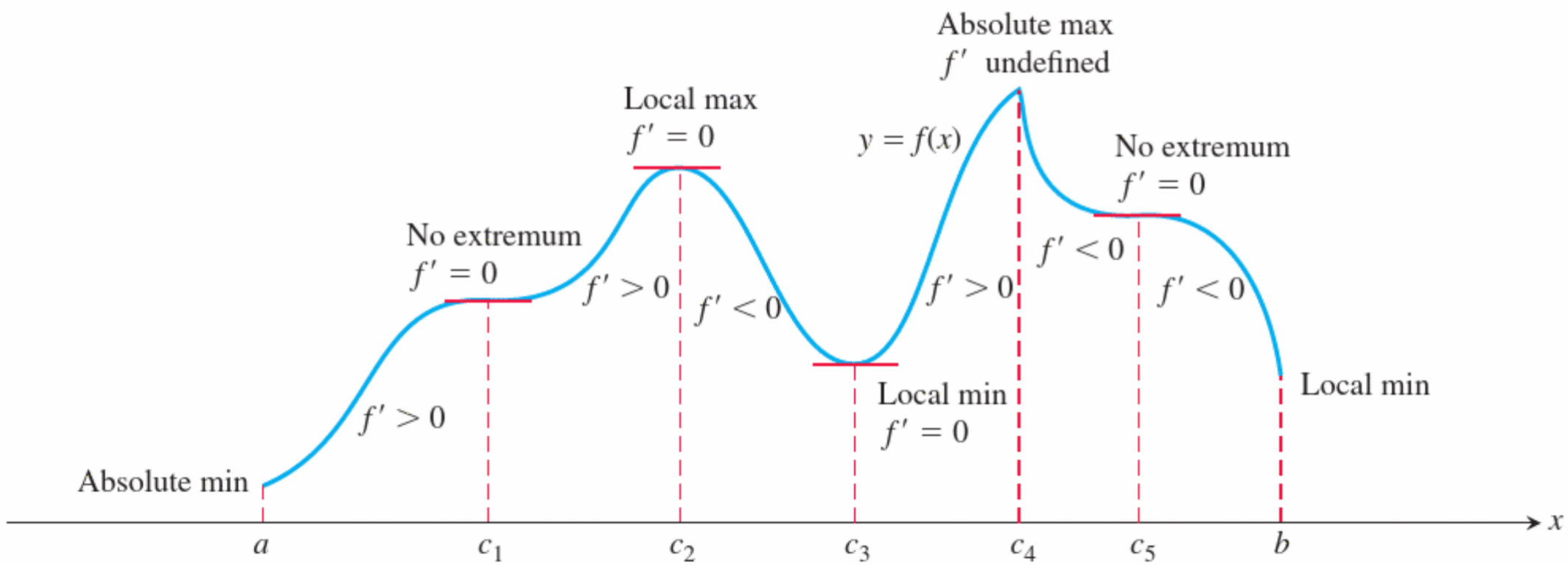


FIGURE 4.21 The critical points of a function locate where it is increasing and where it is decreasing. The first derivative changes sign at a critical point where a local extremum occurs.

Image: Thomas Chapter 4

Why are Derivatives so Useful for Optimisation?

THEOREM 2—The First Derivative Theorem for Local Extreme Values If f has a local maximum or minimum value at an interior point c of its domain, and if f' is defined at c, then

$$f'(c) = 0.$$

First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f, and that f is differentiable at every point in some interval containing c except possibly at c itself. Moving across this interval from left to right,

1. if f' changes from negative to positive at c, then f has a local minimum at c;
2. if f' changes from positive to negative at c, then f has a local maximum at c;
3. if f' does not change sign at c (that is, f' is positive on both sides of c or negative on both sides), then f has no local extremum at c.

Image: Thomas Chapter 4

But it gets even better..!

Second Derivative

Simply the derivative of the 1st derivative

$$f''(x) \equiv \frac{d^2}{dx^2}y \equiv \frac{d^2y}{dx^2}$$

where

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Example

- Let $y = x^7 + 3x^5 + x^{\frac{3}{2}}$
- Then $\frac{dy}{dx} = 7x^6 + 15x^4 + \frac{3}{2}x^{\frac{1}{2}}$
- and $\frac{d^2y}{dx^2} = 42x^5 + 60x^3 + \frac{3}{4}x^{-\frac{1}{2}}$

Q. What is $\frac{d^2}{dx^2}(e^x)$?

Why are Second Derivatives so Useful for Optimisation?

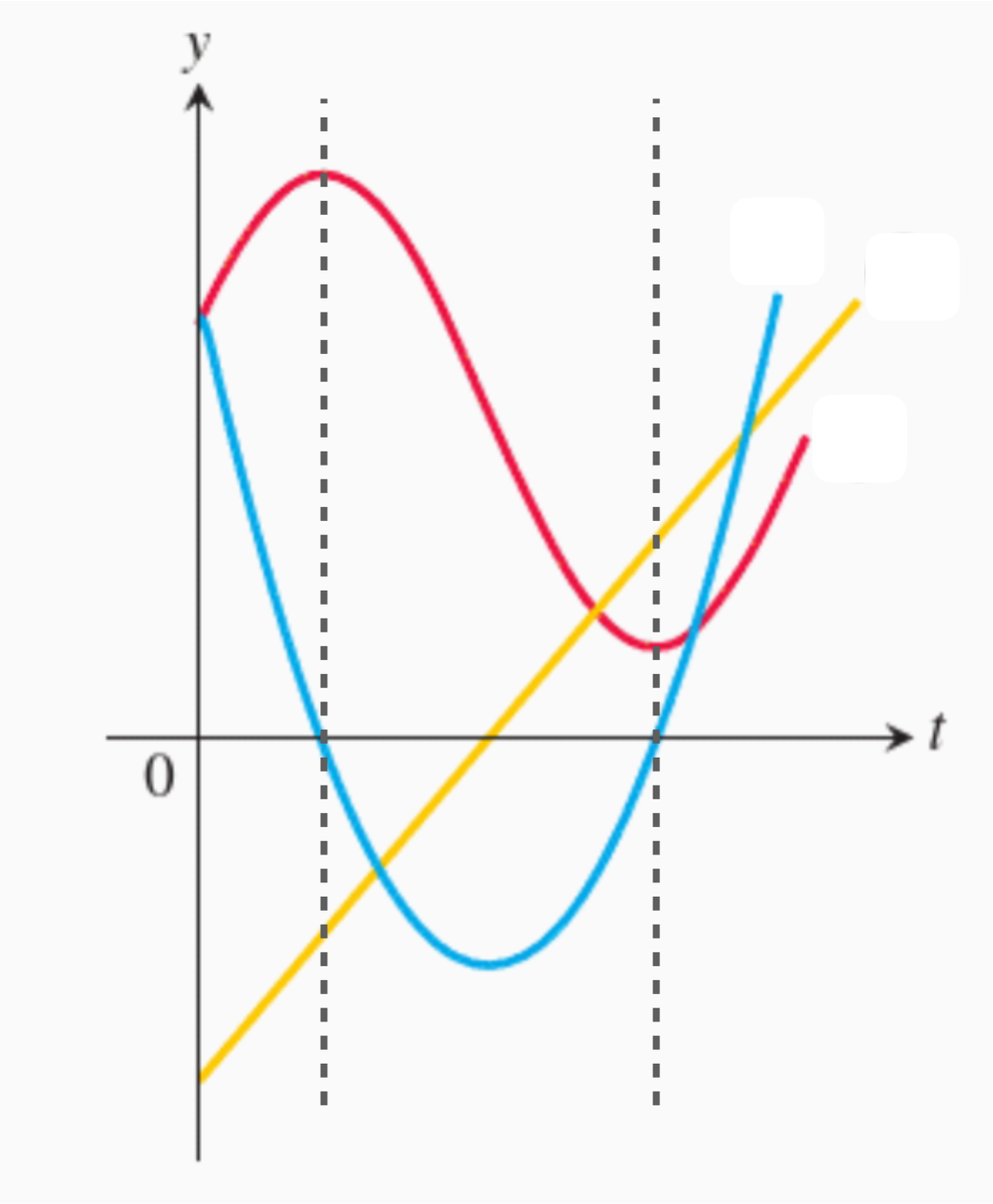


Image: Thomas Chapter 3

Why are Second Derivatives so Useful for Optimisation?

The Second Derivative Test for Concavity

Let $y = f(x)$ be twice-differentiable on an interval I.

1. If $f'' > 0$ on I, the graph of f over I is concave up.
2. If $f'' < 0$ on I, the graph of f over I is concave down.

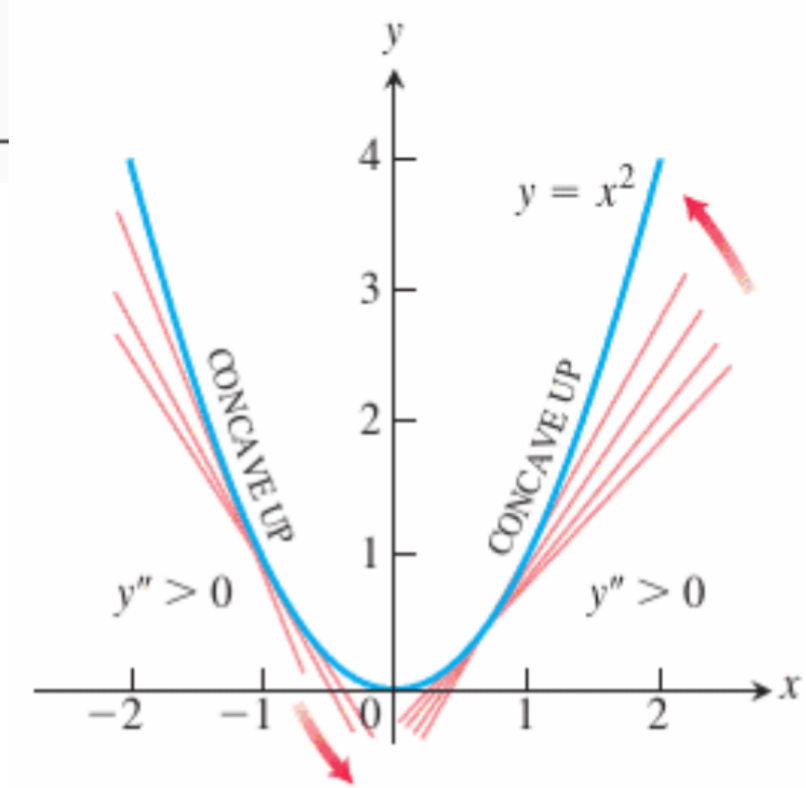


FIGURE 4.25 The graph of $f(x) = x^2$ is concave up on every interval (Example 1b).

Image: Thomas Chapter 4

Why are Second Derivatives so Useful for Optimisation?

THEOREM 5—Second Derivative Test for Local Extrema Suppose f'' is continuous on an open interval that contains $x = c$.

1. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.
2. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.

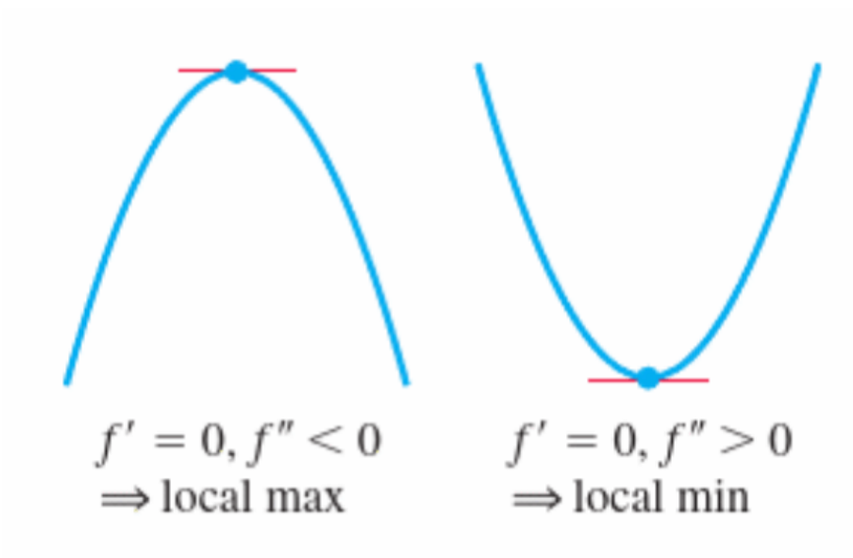


Image: Thomas Chapter 4

Part 2: Vector Calculus

Multivariate Differentiation

References

- *Thomas' Calculus in SI Units*, 13th Ed., Pearson 2016
- Many other books on Vector/Multivariate Calculus

Vectors

magnitude & direction

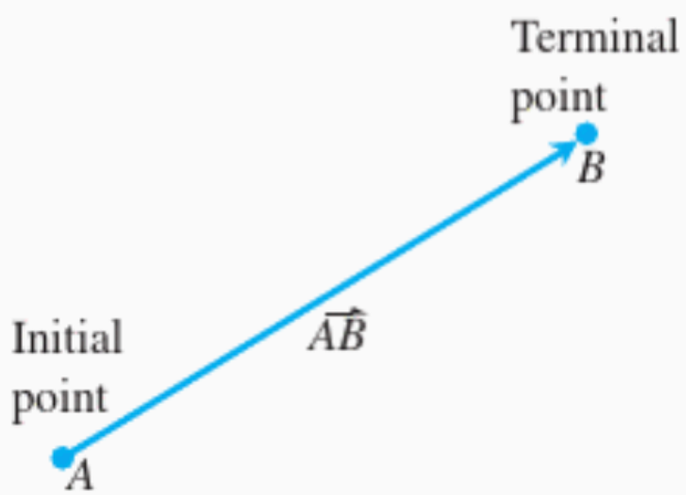


FIGURE 12.7 The directed line segment $\overrightarrow{AB}$ is called a vector.

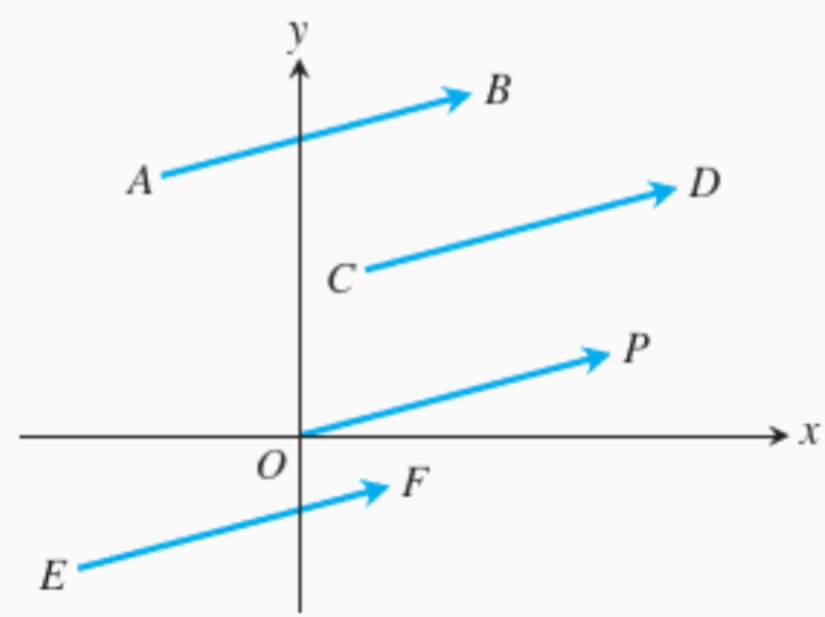


FIGURE 12.9 The four arrows in the plane (directed line segments) shown here have the same length and direction. They therefore represent the same vector, and we write $\overrightarrow{AB} = \overrightarrow{CD} = \overrightarrow{OP} = \overrightarrow{EF}$.

Images: *Thomas' Calculus*, educational use only, not for distribution

Euclidean Distance

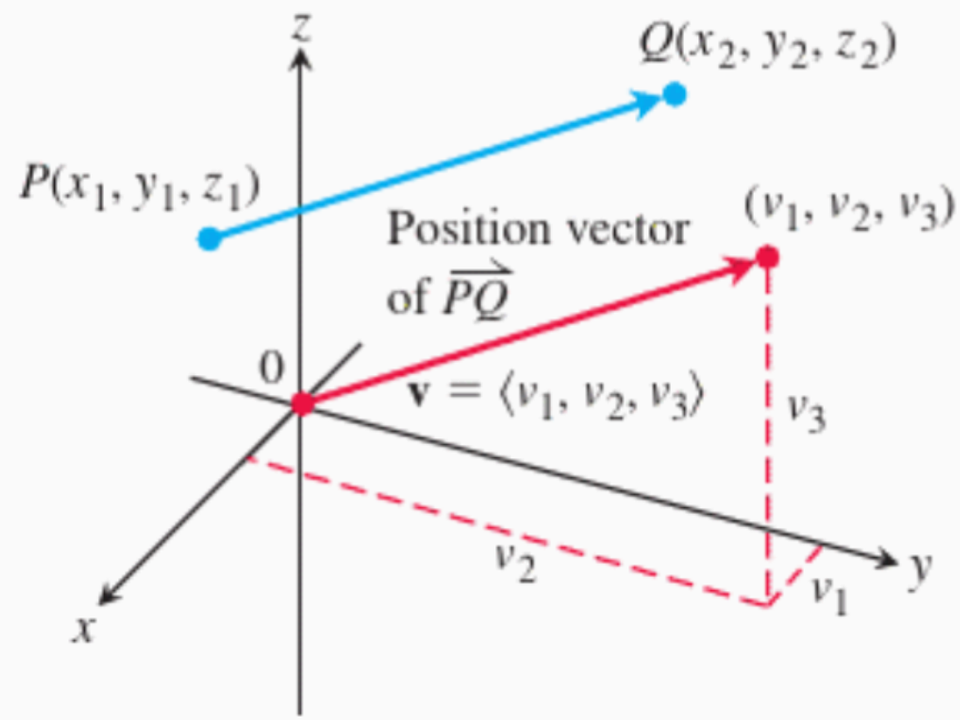


FIGURE 12.10 A vector $\overrightarrow{PQ}$ in standard position has its initial point at the origin. The directed line segments $\overrightarrow{PQ}$ and $\mathbf{v}$ are parallel and have the same length.

magnitude a.k.a. **Euclidean distance** a.k.a. L^2 **norm**

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Vector Addition

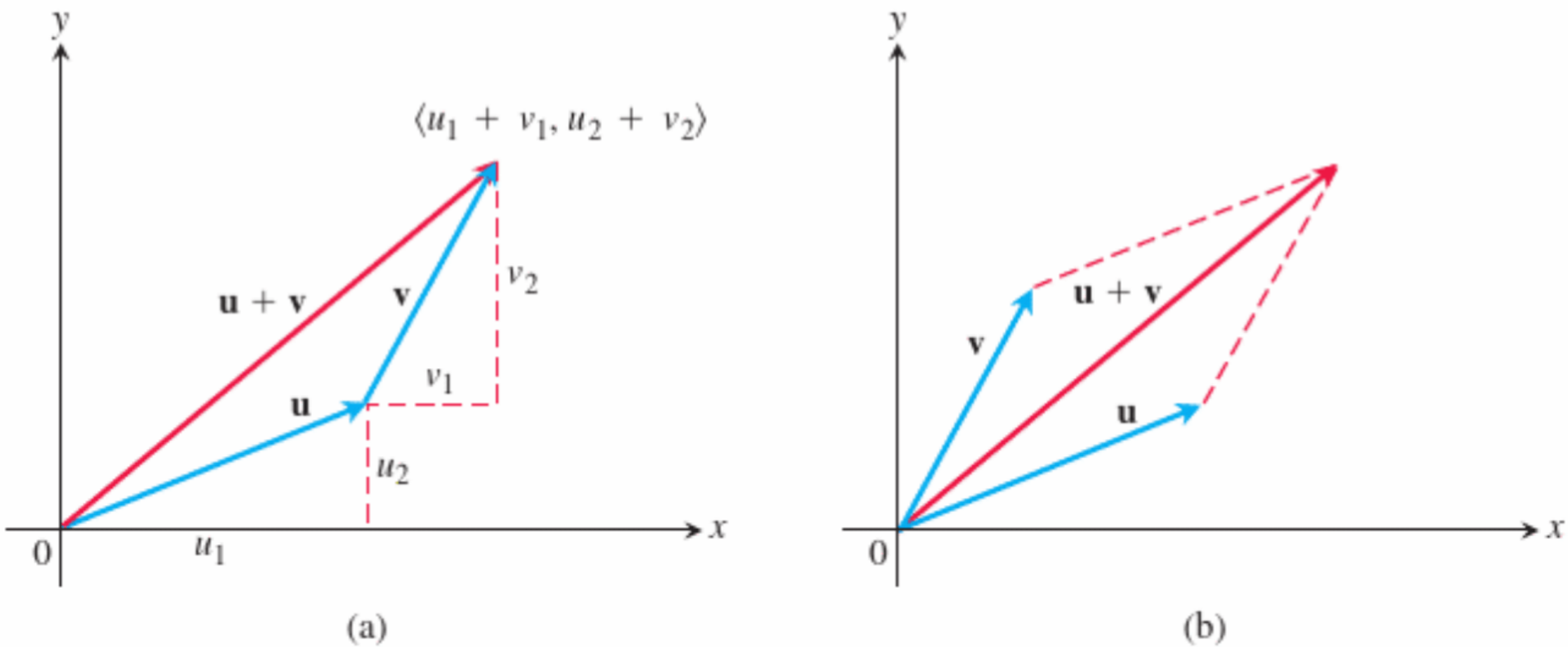


FIGURE 12.12 (a) Geometric interpretation of the vector sum. (b) The parallelogram law of vector addition in which both vectors are in standard position.

Subtraction (Difference)

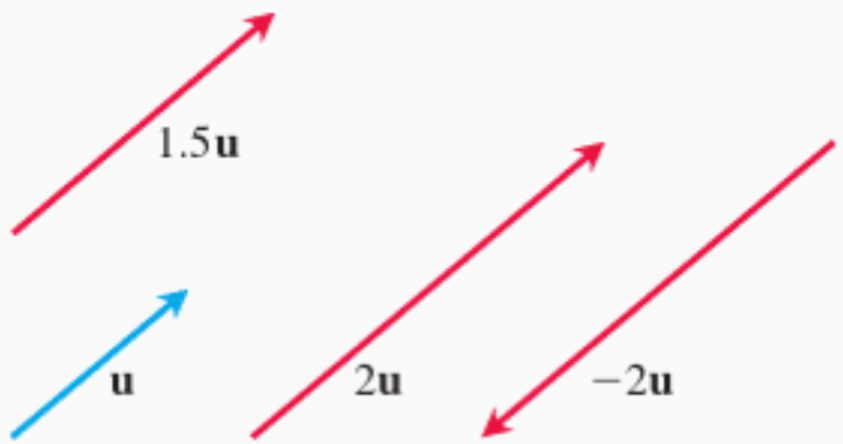


FIGURE 12.13 Scalar multiples of **u**.

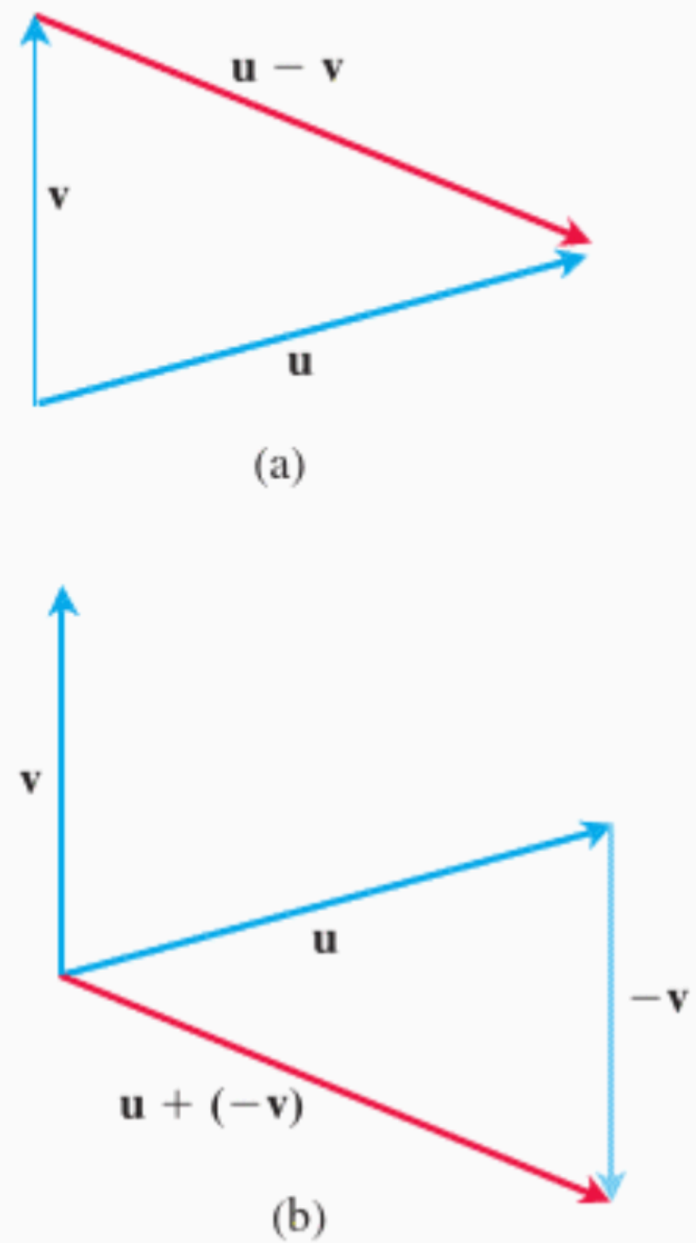


FIGURE 12.14 (a) The vector $\mathbf{u} - \mathbf{v}$, when added to $\mathbf{v}$, gives $\mathbf{u}$. (b) $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$.

Three Kinds of Multiplication

Dot Product

Also known as *inner product* or *scalar product* (result is a scalar)

$$
\begin{aligned}
\vec{v} \bullet \vec{w} &= \sum_{i=1}^{n} v_1 w_1 + v_2 w_2 + \cdots + v_n w_n \\
&= \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \\
&= \vec{v}^{\top} \vec{w}
\end{aligned}
$$

Hence

$$\vec{a} \bullet \vec{a} = \|\vec{a}\|^2 \quad \text{or} \quad \|\vec{a}\| = \sqrt{\vec{a} \bullet \vec{a}}$$

Tensor Product

Also known as *outer product*

$$\vec{v} \otimes \vec{w} = \begin{bmatrix} v_1 w_1 & v_1 w_2 & \cdots & v_1 w_n \\ v_2 w_1 & v_2 w_2 & \cdots & v_2 w_n \\ & & \vdots & \\ v_m w_1 & v_m w_2 & \cdots & v_m w_n \end{bmatrix}$$

$$= \vec{v}\,\vec{w}^\top$$

Hadamard Product

Also known as *element wise product* or *Schur product*

$$\vec{v} \odot \vec{w} = \begin{bmatrix} v_1 w_1 \\ v_2 w_2 \\ \vdots \\ v_n w_n \end{bmatrix}$$

Unit Vectors

- ***Standard unit vectors***

$$\vec{i} = [1,0,0], \quad \vec{j} = [0,1,0], \quad \vec{k} = [0,0,1]$$

- Any vector can be written as a *linear combination of the standard unit vectors*

$$\vec{v} = [v_1, v_2, v_3]$$
$$= v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$$

We will switch between these two representations as convenient

As we will be considering **more than 3 dimensions** we will represent the standard unit (or *elementary*) vectors in $\mathbb{R}^n$ as

$$\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n$$

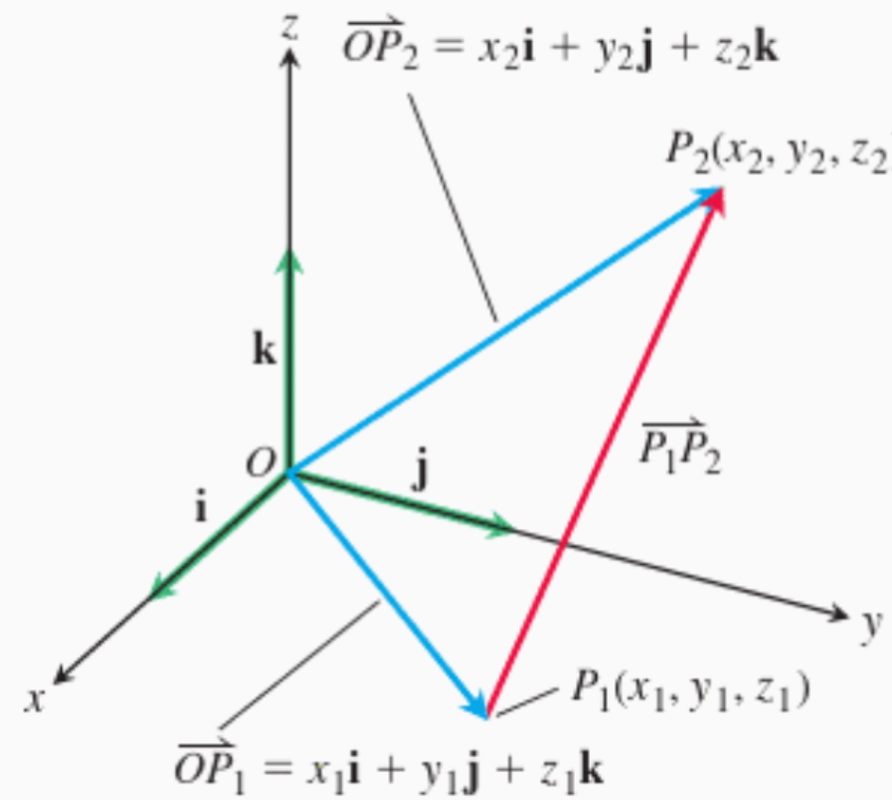


FIGURE 12.15 The vector from P_1 to P_2 is $\overrightarrow{P_1P_2} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$.

Vector Fields

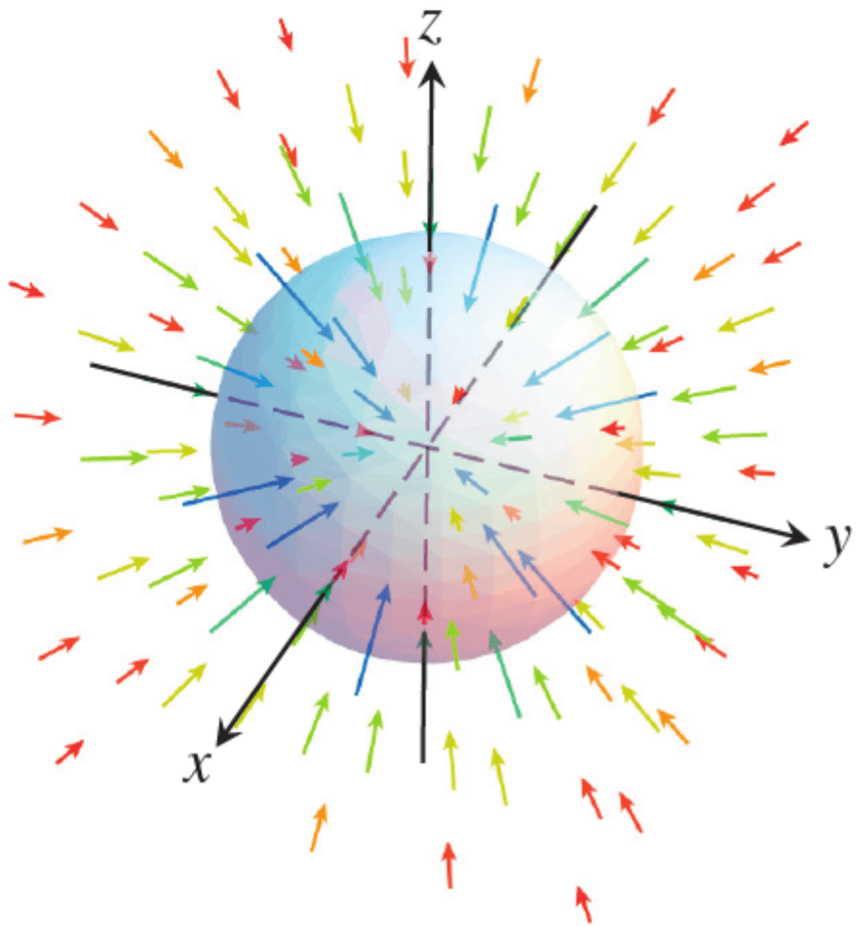
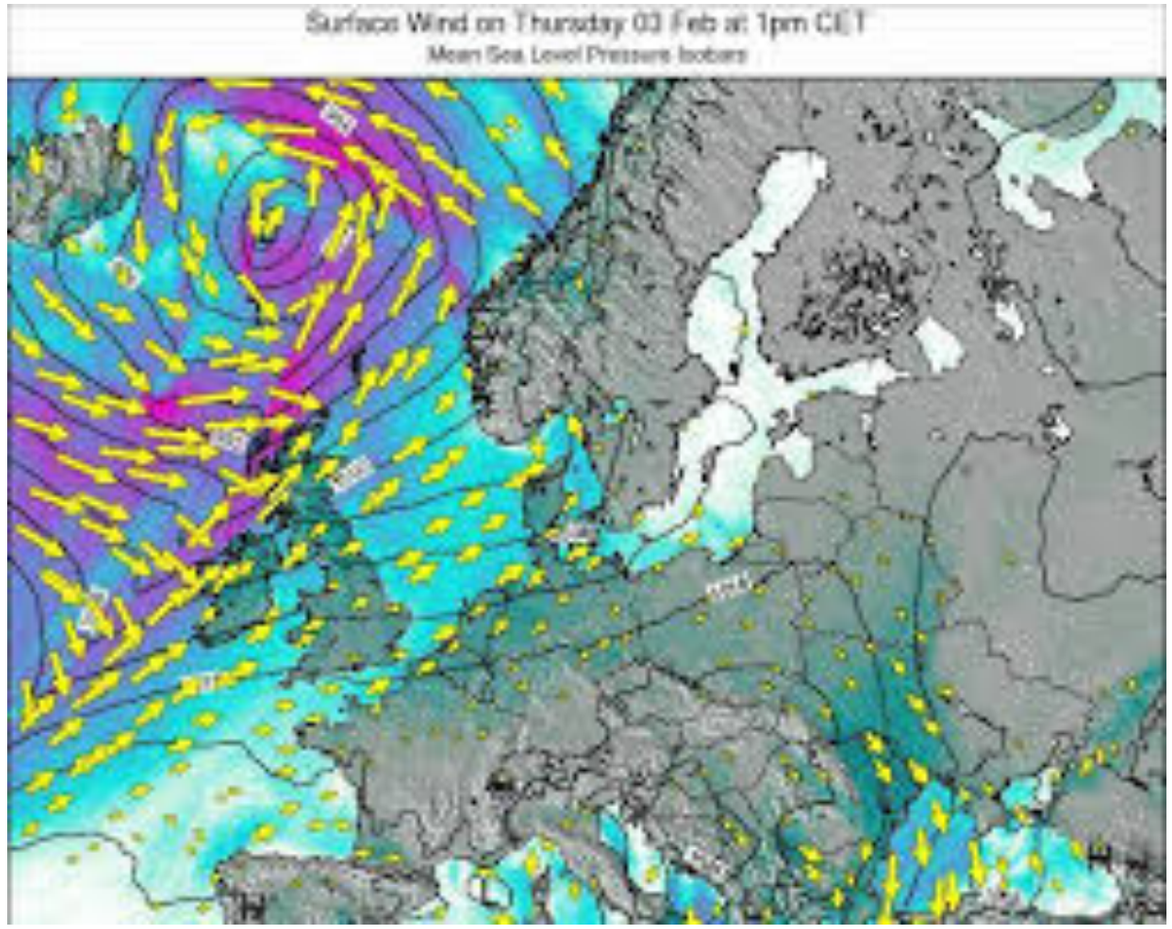


FIGURE 16.8 Vectors in a gravitational field point toward the center of mass that gives the source of the field.



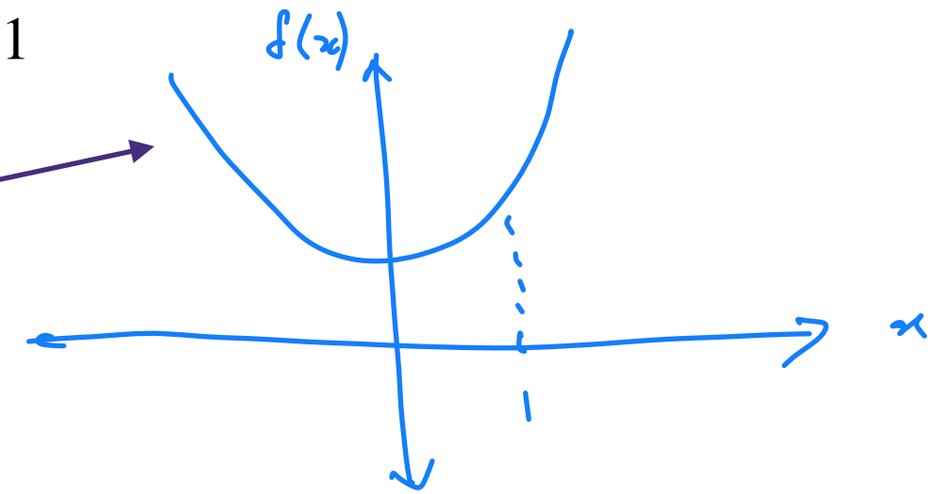
Winds as a vector field

☞ **Back to gradients...!**

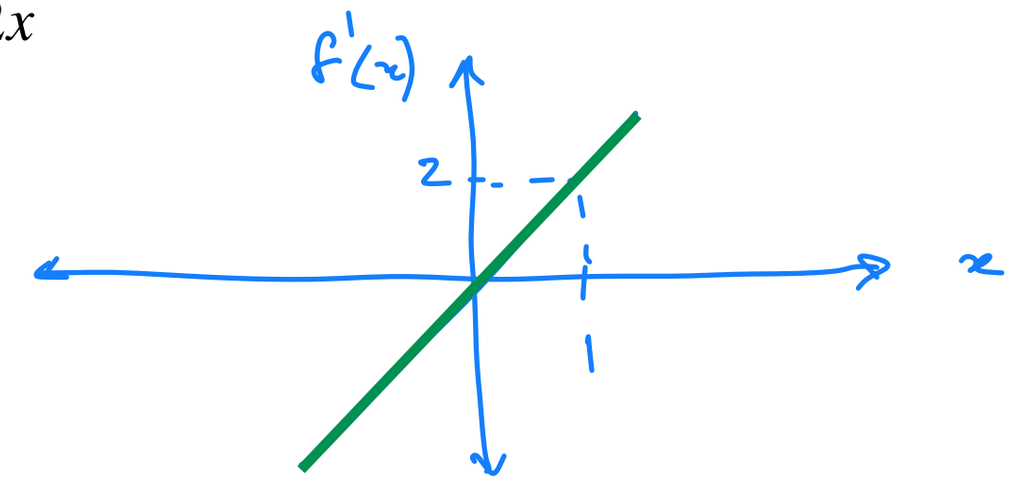
Another way of thinking about Gradient

$$f(x) = x^2 + 1$$

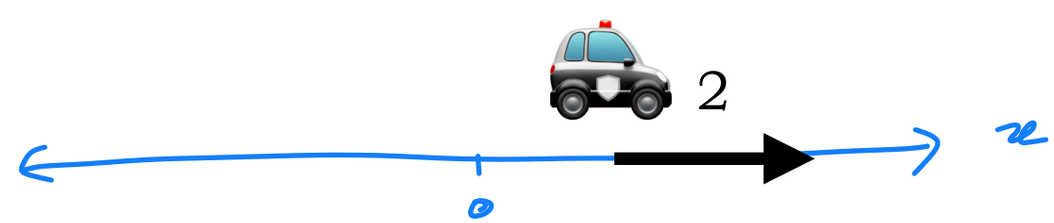
eg. f(x) = car speed



$$f'(x) = 2x$$



- When $x = 1$, the gradient has ***magnitude*** 2 and is in the **direction** of the positive x axis
- Can think of this as a ***vector*** in $\mathbb{R}^1$

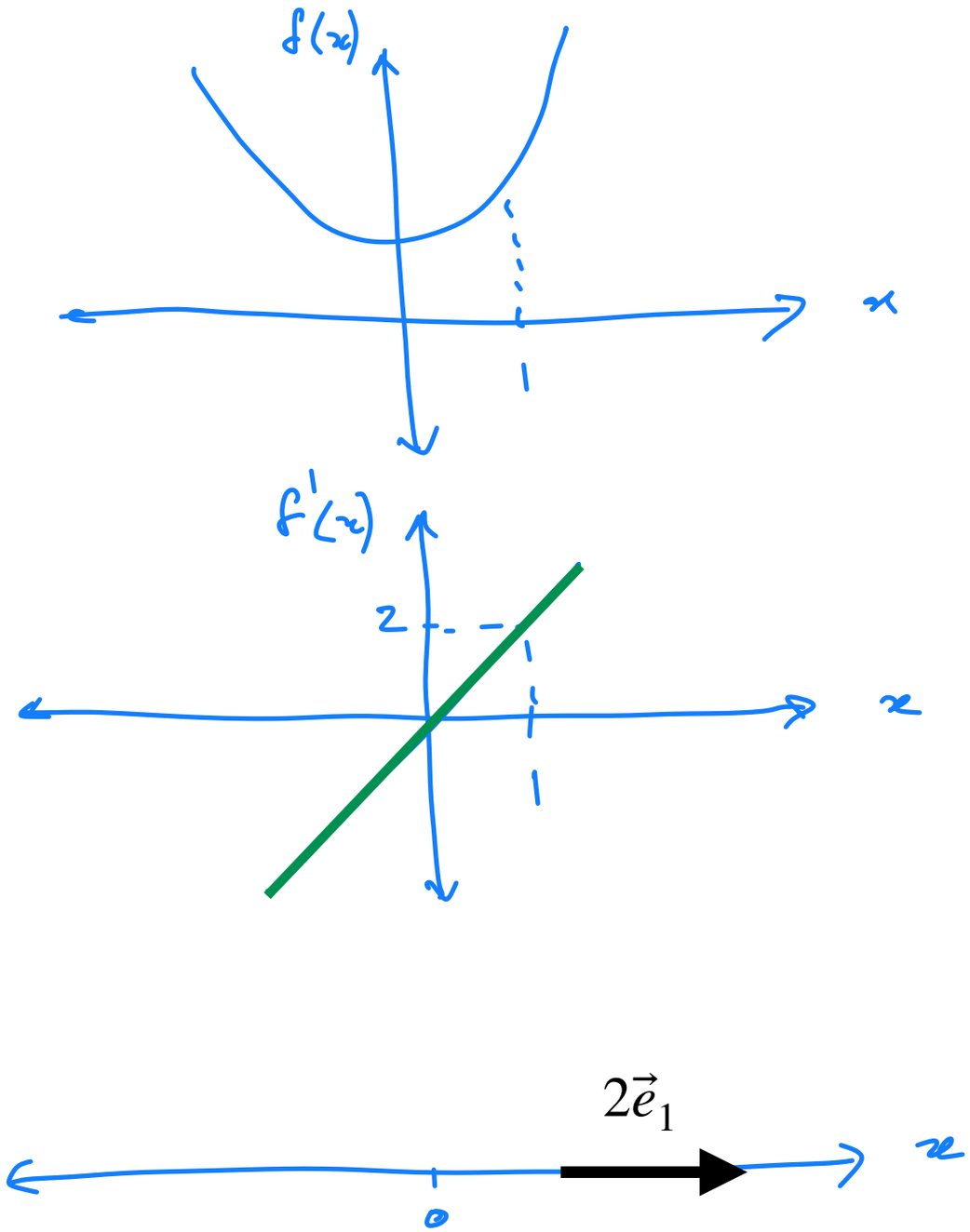


(Actually acceleration vector.)

Another way of thinking about Gradient

- Let $\vec{e}_1$ be a unit (or "elementary") vector in $\mathbb{R}^1$
  - magnitude 1
  - +ve x direction
- Gradient at $x = 1$ can be represented by the vector

  $2\vec{e}_1$

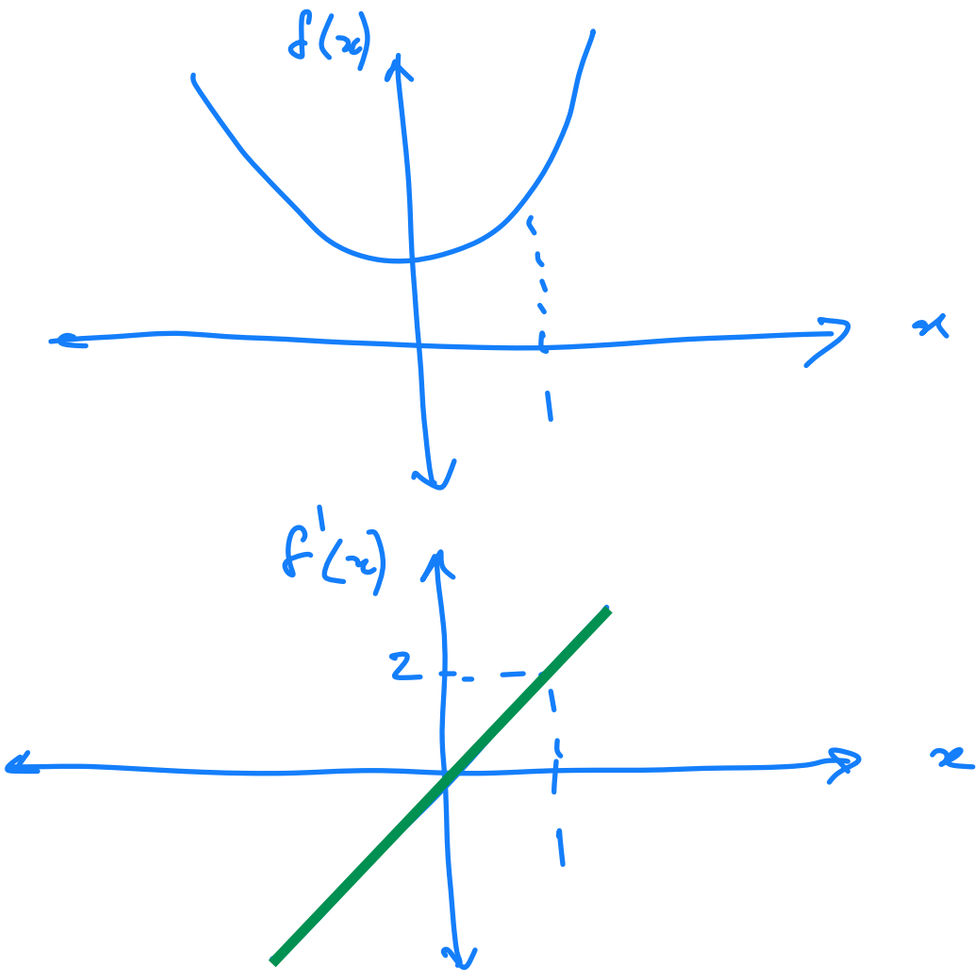


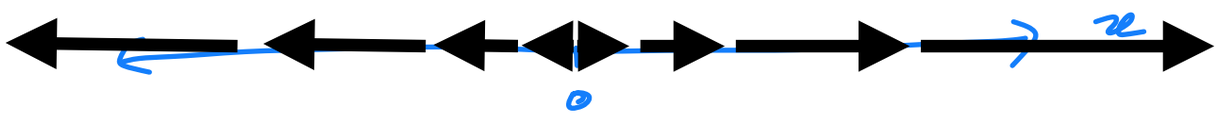
Another way of thinking about Gradient

- In fact, for any ***scalar*** value x, the gradient is given by

$$2x\vec{e}_1$$

- There are an infinite number of these vectors
- Constitute a ***vector field***

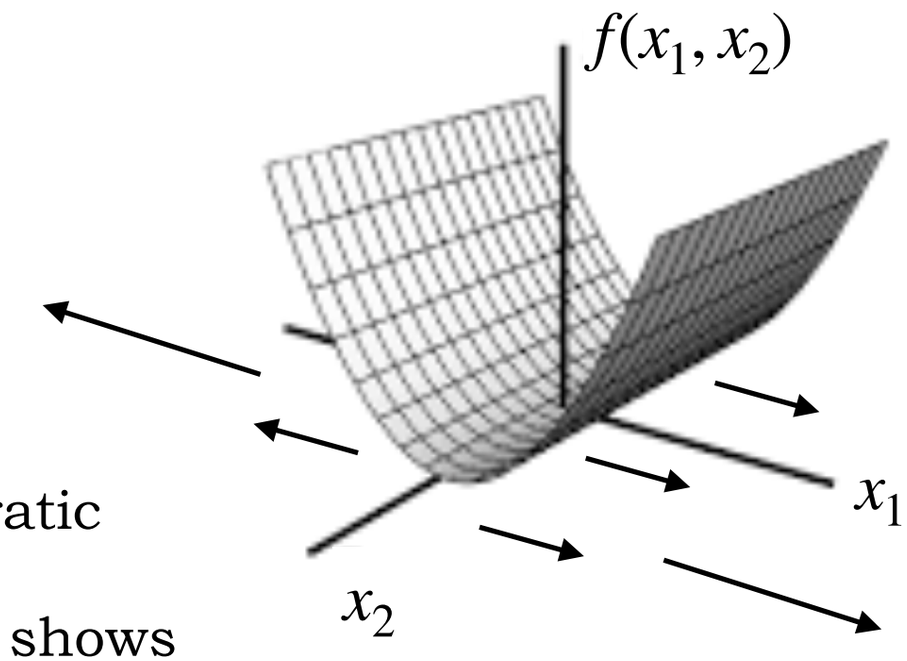




2-dimensions

Consider the function $f(x_1, x_2) = x_1^2$

- "trough" running along the x_2 axis
- doesn't vary with x_2
- looking from the end, you just see a quadratic
- any slice (plane) taken through the x_2 axis shows a quadratic



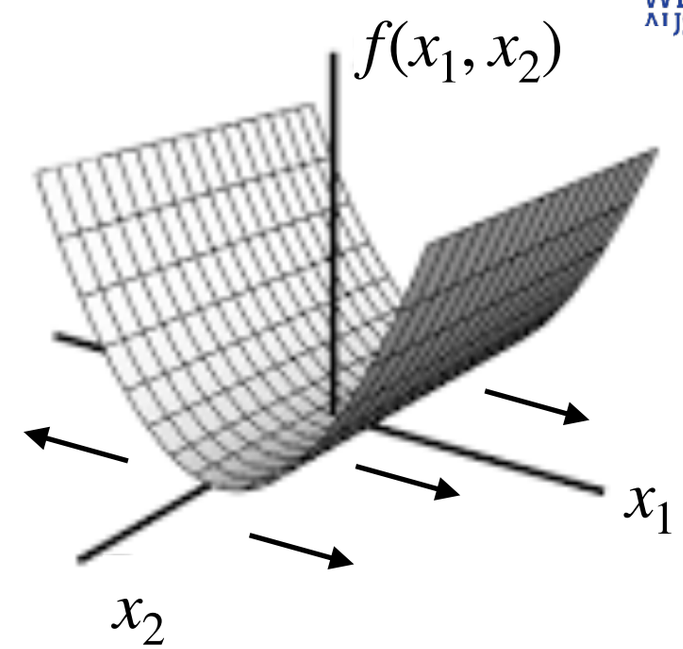
- The gradient also doesn't vary with x_2
  - a plane that is tangential to the surface will be parallel to the x_2 axis
- Gradient can now be represented by a ***vector field*** in 2 dimensions
  - "arrows pointing away from x_2 axis" at all points in the plane
  - defined by ***derivative with respect to*** (wrt) x_1

Partial Derivatives

- Derivative wrt ***one variable*** with ***all other variables fixed (held constant)***

$$f(x_1, x_2) = x_1^2$$

$$\frac{\partial}{\partial x_1} f(x_1, x_2) = 2x_1$$



- Exercise

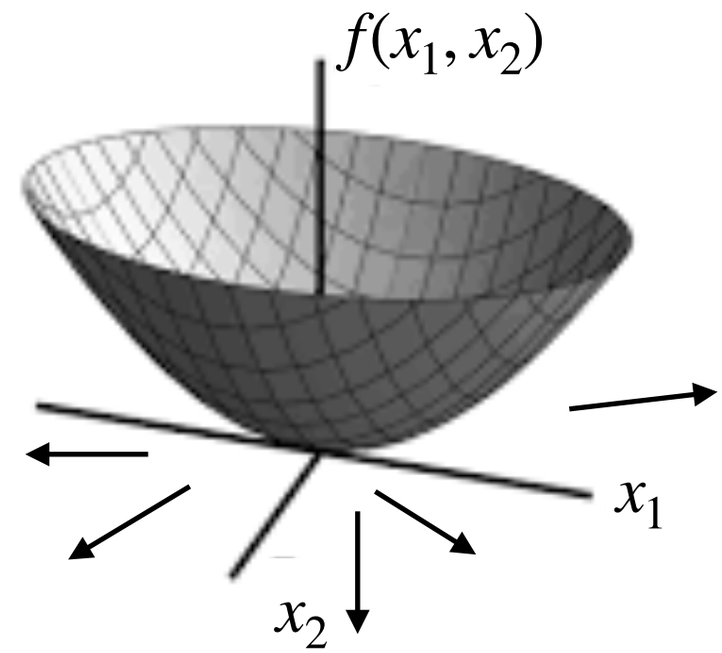
What is $\frac{\partial}{\partial x_2} f(x_1, x_2)$?

  - is it what you expect?

Partial Derivatives

Consider the function $f(x_1, x_2) = x_1^2 + x_2^2$

- "bowl" shape
  - as we "slide" a plane around the surface of the bowl, the direction it faces will change (always away from centre)
  - the direction of maximum slope will also change
  - vector field points away from (0,0)

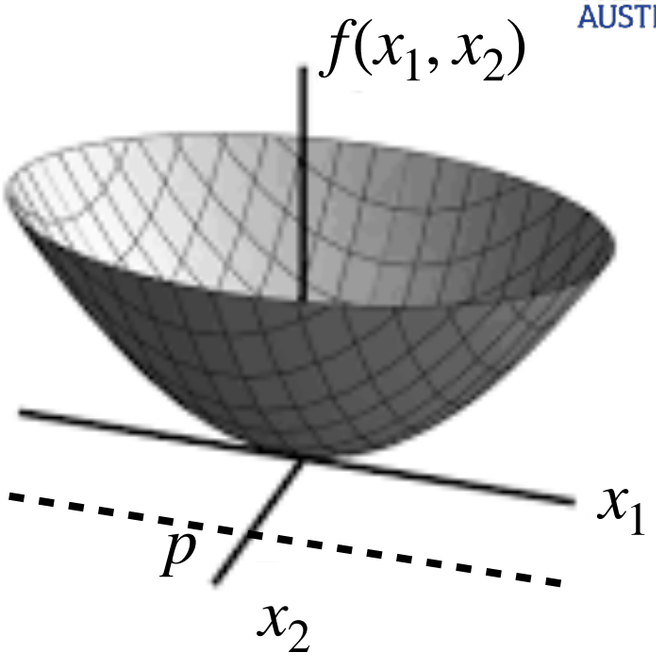


Partial Derivatives

- If we slice the function at any point p along the x_2 axis, the projection of the function on that plane will be a quadratic

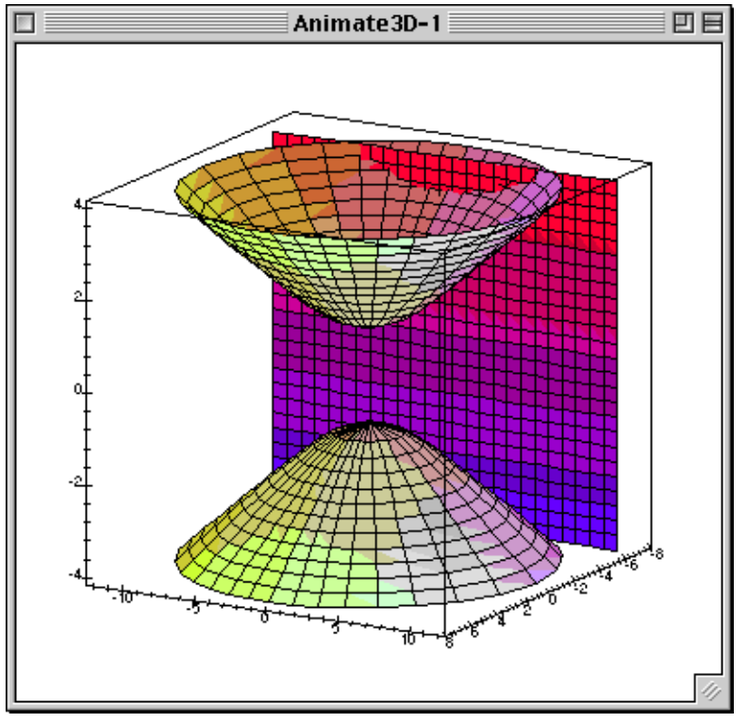
$$x_1^2 + p^2$$

  - contribution to the slope at a point on that function will be $2x_1$ (independent of x_2)
  - as vector, $2x_1\overrightarrow{e_1}$



- Can find this by partial differentiation wrt x_1

$$\frac{\partial}{\partial x_1}(x_1^2 + x_2^2) = 2x_1$$



- This slice is a projection into 1-d. What if we don't slice?

Partial Derivatives

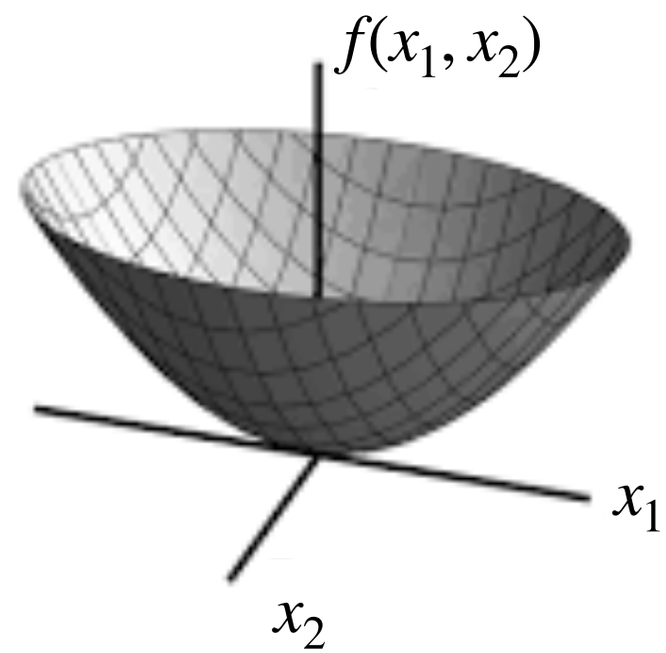
- If we don't slice the function by the plane $x_2 = p$

  then the gradient has a contribution from x_2 as well as x_1

- Contribution from x_2 could be found by an orthogonal slice (fixing x_1) but we know that we can just take the partial derivative

$$\frac{\partial}{\partial x_2}(x_1^2 + x_2^2) = 2x_2$$

- Or as a vector $2x_2\overrightarrow{e_2}$

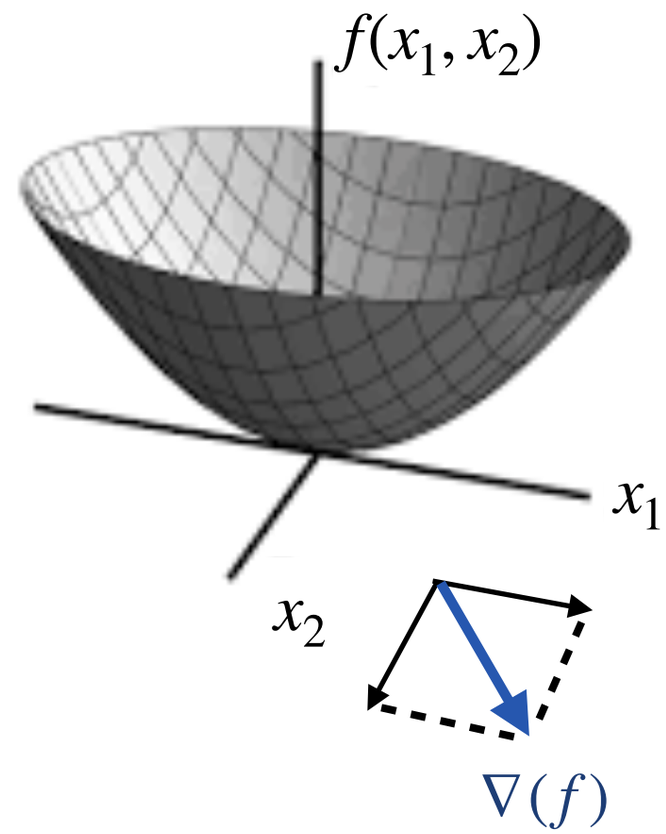


Gradient

- So we have two components to the gradient - how do we get the gradient?
  - ➡ just add the vectors together!

gradient = $2x_1\overrightarrow{e_1} + 2x_2\overrightarrow{e_2}$

- we call this ***grad f*** or $\nabla(f)$



Gradient

In general, if

$$\vec{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$$

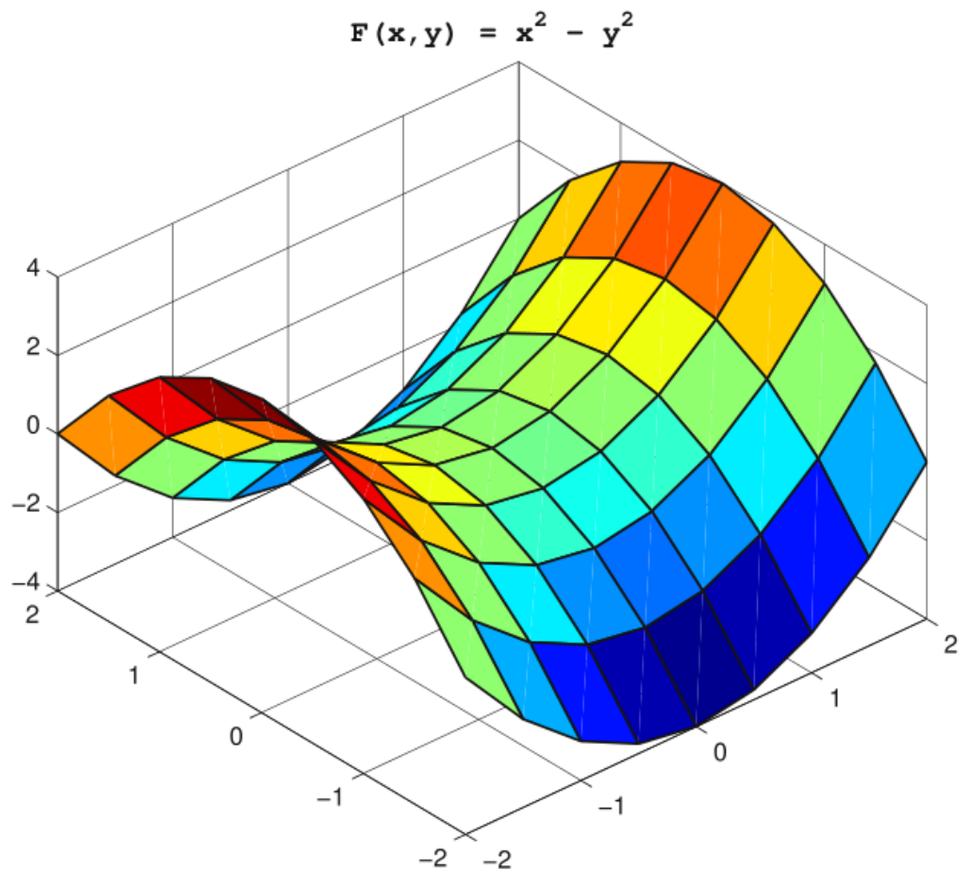
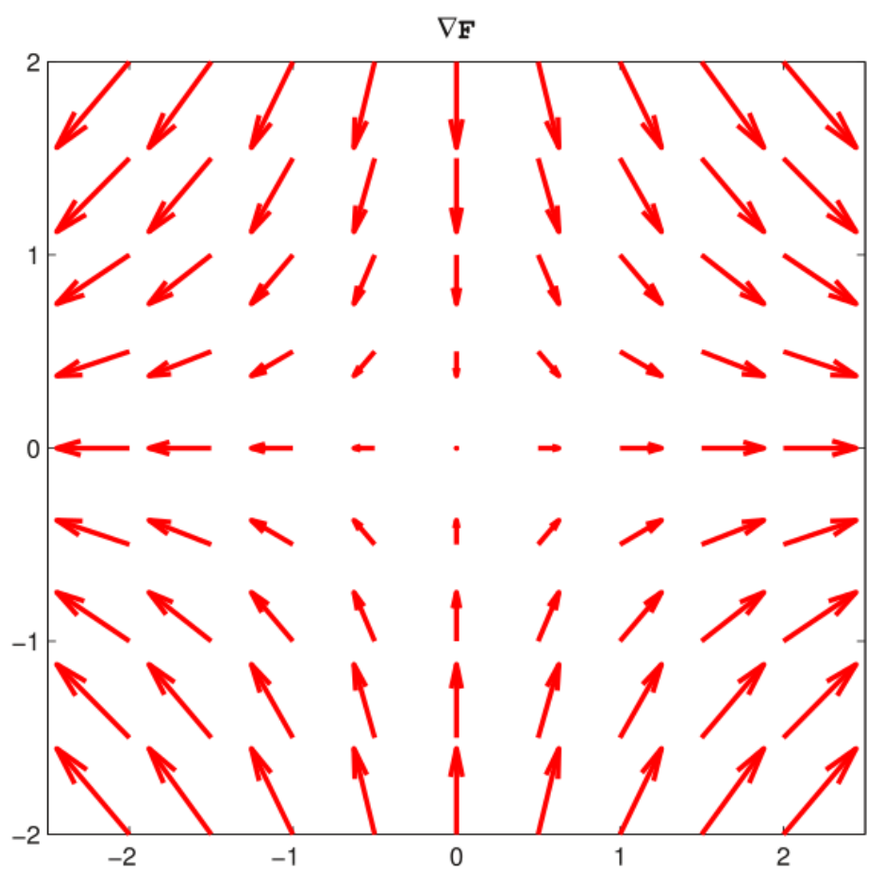
and $f : \mathbb{R}^n \to \mathbb{R}$

then the gradient $\nabla f : \mathbb{R}^n \to \mathbb{R}^n$ is given by

$$\nabla f(\vec{x}) = \frac{\partial f(\vec{x})}{\partial x_1}\overrightarrow{e_1} + \frac{\partial f(\vec{x})}{\partial x_2}\overrightarrow{e_2} + \cdots + \frac{\partial f(\vec{x})}{\partial x_n}\overrightarrow{e_n}$$

or

$$\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f(\vec{x})}{\partial x_1} \\ \frac{\partial f(\vec{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\vec{x})}{\partial x_n} \end{bmatrix}$$

In words...

- gradient of a scalar (differentiable) function f is a vector field
- value at any point is vector whose components are the partial derivatives of f at that point